MAZEN OTHMAN

Data Scientist | Time-Series Anomaly Detection & Operational Decision Support

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Applied mathematician working as a data scientist on live electrical distribution grids: high-frequency time-series anomaly detection, Databricks SQL pipelines, and triage logic that utility dispatch teams act on within minutes. Detected a line break in a dense-vegetation, high fire-risk corridor and reported it to the utility in **under 6 minutes**.

EDUCATION

Imperial College London — London, UK

MSc Data Science and Machine Learning — Expected 2027

Thesis: probabilistic hazard detection and localization on radial distribution feeders — belief propagation over tree-structured graphs with heat-kernel spatial smoothing, benchmarked on IEEE 13/34/37/123-bus test feeders.

Coursework projects:

- User-conversion classifier: logistic regression and KNN with mRMR feature selection and Bayesian hyperparameter search.
- Gene-expression regression: LightGBM and GAMs with GroupKFold cross-validation to generalize to unseen laboratories.
- EDA and clustering in R: missing-data (MCAR) testing, robust scaling, VIF analysis, k-means under two distance definitions.
- Dimensionality reduction: PCA and NMF image denoising, ICA versus PCA for source separation, classical MDS.

Modules: Supervised & Unsupervised Learning, Bayesian Methods and Computation, Exploratory Data Analytics.

In progress: Unstructured Data Analysis (text, images, networks), Learning Agents. **Next:** Deep Learning, Big Data (PySpark).

The American University in Cairo — Cairo, Egypt

BSc Mathematics and BSc Physics — **Summa cum laude · GPA 3.96 / 4.00**

Physics thesis: gradient-descent optimization of nano-antenna designs in the visible spectrum; teaching assistant, 2022–2024.

EXPERIENCE

Gridware Inc. — Remote · Contractor

Data Analyst, Tier 3 — Jun 2025 – Present

- Triage high-frequency telemetry from deployed grid sensors in real time, routing hazard escalations to utility dispatch inside **30 minutes**, and tune detection thresholds against false-alarm versus missed-fault cost.
- Built Databricks SQL pipelines that classify operational event data and feed Tableau dashboards used for live time-to-report tracking, cutting reporting turnaround by **20%**.
- One of **five** internal subject-matter experts on grid anomaly detection; mentor new analysts on shift and act as the escalation point to utility dispatch during live incidents.
- Own workforce planning and scheduling for **40+** analysts across 24/7 coverage, reallocating headcount to clear bottlenecks.
- Work with engineers on label quality and feature consistency, so predictive models train on operational ground truth.

Operations Analyst — Apr 2024 – Jun 2025

- Co-designed Python anomaly-detection logic for real-time grid data streams, improving signal-to-noise ratio and cutting unnecessary escalations by **50%**.
- Monitored distribution-grid time series on shift, escalating critical events to the on-call analyst within **5 minutes**.

The American University in Cairo — Cairo, Egypt

Research Assistant — Jan 2022 – Dec 2022

- Worked with the STAR collaboration at Brookhaven National Laboratory on isolating non-flow contributions to the elliptic flow parameter; benchmarked terabytes of collision data using Pythia Monte Carlo simulations and ROOT.

PROJECTS

European Power Market Forecasting Pipeline

- Built an end-to-end pipeline forecasting next-day hourly day-ahead electricity prices for the German (DE-LU) market from energy-charts.info and Open-Meteo data.
- Engineered 24 features across calendar encodings, price lags, weather fundamentals, and interaction terms; validated LightGBM against a seasonal-naive baseline using walk-forward cross-validation for a **35% MAE improvement**.
- Added an Anthropic Claude API layer that writes daily market commentary from the structured forecast, with deterministic fallback and full prompt/output logging for auditability.

Operational Efficiency Analysis · *Gridware Inc.*

- Self-initiated analysis of 16 weeks of operational data: designed an event-bracket taxonomy to classify reported cases and identified the event categories behind **67%** of late-reporting cases, which fed targeted process changes.

SKILLS

Languages Python, SQL, R, Bash/Shell · **Spoken** English, Arabic

ML & Statistics Time-series analysis, anomaly detection, Bayesian inference, gradient boosting (LightGBM), GAMs, clustering, feature selection, walk-forward and grouped cross-validation, mathematical optimization

Data & Tools Databricks, PySpark, PostgreSQL, Tableau, Git, pandas, NumPy, scikit-learn, PyTorch, Matplotlib, LLM API integration

Certifications IBM Data Analyst Professional Certificate (IBM, 2024)